560. Subarray Sum Equals K Date:

Given an array of Integers nums and and integer K return the total number of SubArrays whose Sum equals to K.

* SubArray is a contignous non-empty sequence of elements within an array.

Example1:-

Input: nums = [1,1,1], K = 2

output : 2

Explanation nums[0] + nums[1] = 2
nums[1] + nums[2] = 2 → 2 sub Array

Example 2:-

Input : nums = [1,2,3], K = 3

output : 2

Explanation:- nums[0] + nums[1] = $\boxed{3}$
num[2] = $\boxed{3}$ → total 2 SubA

* we can solve it by Brute froce method like we can find All SubArrays sum than If it equal to K Count increace by 1 but the time comp is $\simeq O(N^3)$ or with better solution $\simeq O(N^2)$

* Better solution same but with two loop like nums [1][1][1]
→ 1st, 2nd, 3rd, 4rth SubArrays

* we can find their sum by two loop like suppose [1] → 1st subArrays sum = 1
[1, 1] → 2nd subArray sum = 2

we have to Add "1 index value = 1 in sum "0 index value" are present in sum.

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* optimal Solution :-

nums = [1 | 2 | 3 | -3 | 1 | 1 | 1 | 4 | 2 | -3]   K = 3

0 1 2 3 4 5 6 7 8 9

* If we find sum of limited Range
=> suppose on 6 Index

[1 | 2 | 3 | -3 | 1 | 1 | 1]

0 1 2 3 4 5 6

1 + 2 + 3 + (-3) + 1 + 1 + 1 = 6

[1 | 2 | 3 | -3 | 1 | 1 | 1] -> total sum = 6

equal K -> num[4] + num[5] + num[6] = K

PrefixSum = nums[0] + nums[1] + num[2] + num[3] = total sum − K

right

* If we count [total sum − K] mean we can find sum of SubArray = K

* why count ? May be [total sum − K] are more than 1 these are prefixSum

[1 | 2 | 3 | -3 | 1 | 1 | 1] -> total Sum = 6

num[2] + nums[3] + nums[4] + num[5] + nums[6] = K

PrefixSum = nums[0] + nums[1] = total Sum − K

* see total sum = 6 with same number of Index but number prefix (total sum − K) = 2

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* which datastructure we will use and why?

⇒ we will use Hashmap b/c we need to count of prefixSum and prefixSum may dublicate so we will set (keys, values) ⟶ keys = prefix sum, values = count.

like in back side ~~exp~~ example prefixSum=3 their count =2 mean dublicate prefixSum just we have to increase the count b/c keys can not be dublicate.

⇒ If we fount keys have values mean there are number of SubArray.

Dry Run Example nums=[1,2,3], K=3

steps	elements	currsum	currsum-K	HashMap	Count
init	—	—	—	{0:1}	0
1	1	1	1−2	{0:1, 1:1}	0
2	2	3	0	{0:1, 1:1, 3:1}	0+1=1
3	3	6		{0:1, 1:1, 3:1, 6:1}	1+1=2

→ this is b/c may be input is one number like {3}

K = 3

currentSum = 3

currentSum − K = $\boxed{0}$ → already in Hash map.